



Health & Safety

Standard Confined spaces



Reference

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The standards presented in this document aim to provide a practical guide to define the means and organisation to be deployed to effectively manage confined spaces.

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1. Definition of a confined space

A confined space is defined as a partially or totally enclosed space with the following characteristics:

- Is not designed or configured for continuous occupancy
- Workers may experience limited mobility due to narrow passages or cramped conditions.
- Limited entry and exit, typically with restricted openings

When entering these spaces, workers may be exposed to a large number of risks, such as atmospheric hazards (low oxygen, toxic gases, flammable gases), entrapment, limited communication, difficult rescue, etc. which need to be controlled.

Examples of confined spaces (non-exhaustive list):

- Manholes, large pipes
- Sewers
- Crawl spaces, pits of all kinds
- Cisterns, silos, reservoirs
- Tanks, reactors in the chemical or nuclear industry, etc.



2. Confined space procedure

- Mandatory specific procedure defining a confined space entry permit
- Workers in confined spaces, including supervisors, workers, works managers, occupational health and safety (OHS) personnel need to be trained and qualified
- Prioritize permanent access to confined spaces, when feasible
- Specific risk assessment on:
 - The environment of the operation (thermal, gas, smoke, ventilation, etc.)
 - The atmospheric control
 - The work environment (signage, access restrictions, etc.)
 - The equipment to be used for the task
 - The number of operators: mandatory presence of a supervisor during confined space entry (located outside)
 - Collective and individual means of protection
 - Means of communication between the outside and the inside
 - Mandatory locking out of different networks or systems
 - Fire permit consideration



CATEC
training system
(France)

Use of Confined
space entry
permit



3. Confined Space Equipment

Prepare the necessary equipment for the intervention, taking into account the work's risk assessment and the environment. and ensure at minimum the following :

- Standard and specific PPE (suit, specific gloves, goggles, etc.)
 - Atmosphere controller: before the entry into a confined space (on 3 levels, top, bottom and middle, respecting the control times identified on the entry permit)
 - During the intervention (individual portable detector)
- Harness and tripod with a winch designed for confined space use
- Ventilation system
- Means of communication
- Individual self-rescuer masks (also called escape mask) or escape hood
- Respiratory protective devices if necessary



Tripod / harness



Self-rescuer
mask



Mechanical
ventilation



Means of
communication



Portable gas detector
(O₂, CO, H₂S, CH₄)

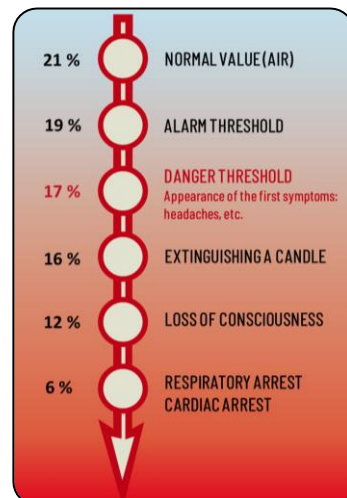
4. Emergency Management / Rescue

The confined space intervention procedure must take into account the rescue/emergency aspect.

- Rescue means (safety lighting, escape hood, defined access, tripod, etc.)
- Identified and trained First aiders ready to intervene if necessary
- Operational emergency procedure, including external emergency communication, etc.)

Maximum number of people present at the same time in a clearly identified confined space:

- According to the conditions on site
- Depending on the operations to be carried out
- Depending on the emergency resources available



Oxygen level in the air and consequences of depletion